

我的提问：

The provided CSV file contains data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb to perform continuous finger tapping. A 10-second segment from this activity was selected for analysis. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement, employing a tracking technique to record the x and y coordinates of the stickers on the thumb and index finger in each frame throughout the 10-second interval.

The objective is to score the subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING based on the selected 10 seconds of continuous finger tapping. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

“Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

0: Normal: No problems.

1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.

2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.

3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.

4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

The provided Python file includes a method for calculating the number of taps. Use this method for the analysis and begin by generating a plot of the data.

To determine the subject's score, analyze the provided data thoroughly and give a detailed motivation for the assigned score based on the MDS-UPDRS finger tapping instruction. Print the score using only one number on the last line of your answer, after the detailed motivation.

Here is the analysis of the finger-tapping data.

The provided Python script was executed to analyze the

879885247_20200803_Hands_R_L.mp4FT.csv file. The analysis first calculated the normalized Euclidean distance between the thumb and index finger for each frame. Then, it identified peaks in this distance signal, which correspond to the moments the fingers are furthest apart (i.e., the "taps").

Analysis Findings

- Data Duration:** The provided CSV file contains 975 frames, which at 240 fps, corresponds to a total duration of **4.06 seconds**.
- Number of Taps:** Within this 4.06-second interval, a total of **11 taps** were detected.
- Tapping Rhythm (Periods):** The time *between* the 11 taps (10 periods) was calculated.
 - Periods (in frames):** [99 92 104 78 106 79 104 78 91 83]
 - Statistics:** The mean period was 91.4 frames, with a standard deviation of 10.83 frames.
 - Interpretation:** A standard deviation of ~12% of the mean, and the visible fluctuation in the period lengths (e.g., from 78 to 106 frames), indicates a clear break from a regular, steady rhythm. This demonstrates multiple hesitations or interruptions in the tapping speed.
- Tapping Amplitude:** The normalized amplitudes (peak heights) for the 11 detected taps were:
 - Amplitudes:** [0.75, 0.94, 0.93, 0.79, 0.73, 1.00, 0.79, 0.72, 0.60, 0.75, 0.80]
 - Interpretation:** The amplitude is highly irregular, not consistently "as big as possible." While it peaks at Tap 6 (1.00), it shows a significant decrement from Tap 6 down to Tap 9 (0.60). This is a clear drop in amplitude occurring in the second half of the sequence.

Scoring Motivation

The plot below visually confirms these findings. The red 'x' markers (taps) are not evenly spaced horizontally (irregular rhythm) and are not at a consistent height (irregular amplitude).

Based on the MDS-UPDRS item 3.4 scoring criteria, we can assess the performance:

- Score 0 (Normal):** The performance is not normal, given the clear irregularities in both rhythm and amplitude.
- Score 1 (Slight):** This score is for "one or two interruptions" or "amplitude decrements near the end." The observed rhythm has more than one or two interruptions, and the main amplitude decrement occurs midway, not just at the end.

- **Score 2 (Mild):** This score is assigned for "Any of the following: a) 3 to 5 interruptions during tapping... c) the amplitude decrements midway in the 10-tap sequence."
 - **Criterion (a):** The tapping period data shows at least 4-5 significant changes in rhythm (e.g., 104 → 78, 78 → 106, 106 → 79, 79 → 104). This matches the "3 to 5 interruptions" criterion.
 - **Criterion (c):** The amplitude data shows a clear and significant decrement from Tap 6 (peak) to Tap 9 (low point). This decrement occurs "midway" or in the second half of the sequence.
- **Score 3 (Moderate):** This is for "> 5 interruptions" or "amplitude decrements starting after the 1st tap." The amplitude *increased* after the 1st tap, so that criterion does not fit. While the rhythm could be argued to have more than 5 interruptions, the criteria for Score 2 are both clearly and strongly met.
- **Score 4 (Severe):** The subject is clearly able to perform the task, so this score is not appropriate.

Conclusion: The performance clearly exhibits both rhythmic interruptions (matching criterion 2a) and a significant amplitude decrement midway through the performance (matching criterion 2c). Therefore, the subject's performance is rated as Mild.